Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

beamer [faculty=science, university=uu,
logo=uu-logo]beamerthemefibeamer [utf8]inputenc [main=english
]babel ragged2e booktabs tabularx tikz calc, shapes,
backgrounds amsmath, amssymb url listings

Presentation Title

Presentation Subtitle

Author's Name

19 Pebruari 2025

Outline for Section 1

Light Frames
- Blind Text
- Structuring Elements
- Numerals and Mathematics
- Figures and Code Listings
- Citations and Bibliography

Jabberwocky

Lewis Carroll

[overlay,remember picture] [anchor=south
east,xshift=-30pt,yshift=35pt] at (current page.south east) ;
'Twas brillig, and the slithy toves
Did gyre and gimble in the wabe;
All mimsy were the borogoves,
And the mome raths outgrabe.

"Beware the Jabberwock, my son!
The jaws that bite, the claws that catch!
Beware the Jubjub bird, and shun
The frumious Bandersnatch!"

Lists and locales

Lorem ipsum dolor sit amet

- Nulla nec lacinia odio. Curabitur urna tellus.
  - Fusce id sodales dolor. Sed id metus dui.
    - Cupio virtus licet mi vel feugiat.

1. Donec porta, risus porttitor egestas scelerisque video.
   1.1 Nunc non ante fringilla, manus potentis cario.
   1.1.1 Pellentesque servus morbi tristique.

Nechť již hříšné saxofony ďáblů rozzvučí síň úděsnými tóny waltzu, tanga a quickstepu! Nezvyčajné kŕdle šťastných figliarskych ďatľov učia pri kótovanom ústí Váhu mĺkveho koňa Waldemara obžierať väčšie kusy exkluzívnej kôry. The quick, brown fox jumps over a lazy dog. DJs flock by when MTV ax quiz prog. “Now fax quiz Jack!”

Text blocks

In plain, example, and alert flavour

This text is highlighted.

A plain block

This is a plain block containing some highlighted text.

An example block

This is an example block containing some highlighted text.

An alert block

This is an alert block containing some highlighted text.

Definitions, theorems, and proofs

All integers divide zero

Definition

$\forall a, b \in Z : a \mid b \iff \exists c \in Z : a \cdot c = b$

Theorem

$\forall a \in Z : a \mid 0$

Proof

$\forall a \in Z : a \cdot 0 = 0$ $\square$

Numerals and Mathematics

Formulae, equations, and expressions

$$1234567890 \quad 1234567890 \quad \hat{x},\ \check{x},\ \tilde{a},\ \bar{a},\ \dot{y},\ \ddot{y} \quad \iint f(x,y,z)\,\mathrm{d}x\mathrm{d}y\mathrm{d}z$$

$$\cfrac{1}{1+\cfrac{1}{2+\cfrac{1}{3+x}}} + \cfrac{1}{1+\frac{1}{2+\frac{1}{3+x}}} \qquad F : \begin{vmatrix} F''_{xx} & F''_{xy} & F'_x \\ F''_{yx} & F''_{yy} & F'_y \\ F'_x & F'_y & 0 \end{vmatrix} = 0$$

$$\iint\limits_{x \in R^2} \langle \mathrm{x}, \mathrm{y} \rangle \,\mathrm{dx} \qquad \overline{\overline{a\alpha^2} + \underline{b\beta} + \overline{\overline{d\delta}}} \qquad]0,1[+ \lceil x \rfloor - \langle x, y \rangle$$

$$e^x \approx 1 + x + x^2/2! + + x^3/3! + x^4/4! \qquad \binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$$

Figures

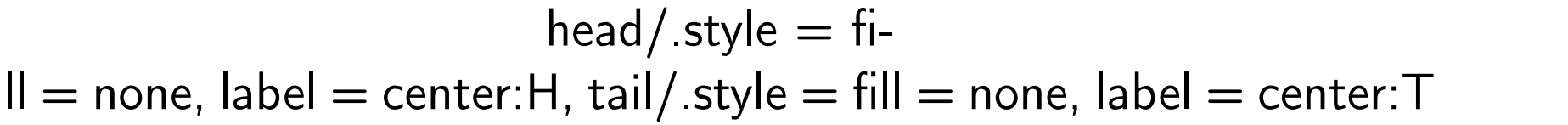
Tables, graphs, and images

Xrrr **Faculty**	**With TEX**	**Total**	**%**
Faculty of Informatics	1 716	2 904	59.09
Faculty of Science	786	5 275	14.90
Faculty of Economics and Administration	64	4 591	1.39
Faculty of Arts	69	10 000	0.69
Faculty of Medicine	8	2 014	0.40
Faculty of Law	15	4 824	0.31
Faculty of Education	19	8 219	0.23
Faculty of Social Studies	12	5 599	0.21
Faculty of Sports Studies	3	2 062	0.15

Tabel: The distribution of theses written using TEX during 2010–15 at MU

Figures

Tables, graphs, and images



Gambar: Tree of probabilities – Flipping a coin[1]

[1] A derivative of a diagram from `texample.net` by cis, CC BY 2.5 licensed

Code listings

An example source code in C

[language=C,tabsize=2] include <stdio.h> include <unistd.h>
include <sys/types.h> include <sys/wait.h>
// This is a comment int main(int argc, char **argv) while (–c >
1 !fork()); sleep(c = atoi(v[c])); printf("wait(0); return 0;

Citations

TEX, LATEX, and Beamer

TEX is a programming language for the typesetting of documents. It was created by Donald Erwin Knuth in the late 1970s and it is documented in *The TEXbook* [1].
In the early 1980s, Leslie Lamport created the initial version of LATEX, a high-level language on top of TEX, which is documented in *LATEX: A Document Preparation System* [2]. There exists a healthy ecosystem of packages that extend the base functionality of LATEX; *The LATEX Companion* [3] acts as a guide through the ecosystem.
In 2003, Till Tantau created the initial version of Beamer, a LATEX package for the creation of presentations. Beamer is documented in the *User's Guide to the Beamer Class* [4].

Bibliography

TEX, LATEX, and Beamer


Donald E. Knuth. *The TEXbook*. Addison-Wesley, 1984.


Leslie Lamport. *LATEX: A Document Preparation System*. Addison-Wesley, 1986.


M. Goossens, F. Mittelbach, and A. Samarin. *The LATEX Companion*. Addison-Wesley, 1994.


Till Tantau. *User's Guide to the Beamer Class Version 3.01*. Available at `http://latex-beamer.sourceforge.net`.


A. Mertz and W. Slough. Edited by B. Beeton and K. Berry. *Beamer by example* In TUGboat, Vol. 26, No. 1., pp. 68-73.